

Ambrogio Maria Bernardelli

CURRICULUM VITAE ET STUDIORUM

Research Positions

01/2025–present Postdoctoral researcher, University of Pavia, Department of Mathematics, for the project *Stochastic optimization for electrical energy storage systems and renewable energy sources*.

Education

Ph.D. in Computational Mathematics and Decision Sciences

University of Pavia and Università della Svizzera Italiana

Cycle: XXXVII.

Dates: 01/2022 – 12/2024.

Thesis title: *Methods for Combinatorial Optimization and Their Applications*.

Supervisor: Prof. Stefano Gualandi.

M.Sc. in Mathematics [LM-40]

University of Pavia

Dates: 10/2019 – 09/2021.

Thesis title: *Aspetti Algebrici e Combinatori della Teoria degli Operad* (Algebraic and Combinatorial Aspects of Operad Theory).

Supervisor: Prof. Alberto Canonaco.

Grade: 110/110 cum laude.

B.Sc. in Mathematics [L-35]

University of Pavia

Dates: 10/2016 – 09/2019.

Thesis title: *Rappresentazioni Lineari di Gruppi Finiti* (Linear Representations of Finite Groups).

Supervisor: Prof. Alberto Canonaco.

Grade: 110/110 cum laude.

Conferences Participation

Conferences and workshops where my works have been presented. Invited talks are labeled with [†].

06/2026 IPCO 2026, Padova, Italy. Poster.

09/2025 The HEXAGON Workshop on power grids, Benevento, Italy[†].

09/2025 ODS 2025, Milan, Italy.

07/2024 IPCO 2024, Wrocław, Poland. Poster.

07/2024 ISMP 2024 (presented by Simone Milanesi), Montréal, Canada[†].

06/2024 EURO 2024 (presented by Simone Milanesi), Copenhagen, Denmark[†].

06/2024 The HEXAGON Workshop on power grids, Bergamo, Italy[†].

05/2024 ISCO 2024, San Cristóbal de La Laguna, Spain.

10/2023 2023 INFORMS Annual Meeting (presented by Stefano Gualandi), Phoenix, Arizona[†].

- 06/2023 LION17, Nice, France. The work *The BeMi Stardust: a Structured Ensemble of Binarized Neural Network* presented here got accepted for the conference proceedings.
- 02/2023 7th AIROYoung Workshop (presented by Simone Milanesi), Milano, Italy.
- 02/2023 7th AIROYoung Workshop, Milano, Italy.
- 01/2023 The Mathematics of Machine Learning (presented by Simone Milanesi), Pisa, Italy.
- 11/2022 Matematica per l'Intelligenza Artificiale e il Machine Learning - Giovani ricercatori, Torino, Italy.
- 09/2022 YAMC 2022, Arenzano, Italy.
- 07/2022 ICCOPT / MOPTA 2022, Lehigh University, Bethlehem, Pennsylvania. The work *Scheduling elective surgeries under uncertainty: a multi-objective stochastic approach* presented here got the second place out of 13 submissions at the 14th AIMMS-MOPTA Optimization Modeling Competition.

Publications

Journal Articles

- [J3] Sosso, A., Bernardelli, A. M., Gualandi, S. (2026). The Cloven Traveling Salesman: Cycle Covers and the Integrality Gap of Small ATSP Instances. *Optimization Letters*.
- [J2] Bernardelli, A. M., Gualandi, S., Milanesi, S., Lau, H. C., Yorke-Smith, N. (2024). Multiobjective Linear Ensembles for Robust and Sparse Training of Few-Bit Neural Network. *INFORMS Journal on Computing*.
- [J1] Bernardelli, A.M., Bonasera, L., Duma, D., Vercesi, E. (2024). Multi-objective stochastic scheduling of inpatient and outpatient surgeries. *Flex Serv Manuf J*.

Conference Proceedings

- [P1] Bernardelli, A.M., Gualandi, S., Lau, H.C., Milanesi, S. (2023). The BeMi Stardust: A Structured Ensemble of Binarized Neural Networks. In: Sellmann, M., Tierney, K. (eds) *Learning and Intelligent Optimization. LION 2023. Lecture Notes in Computer Science*, vol 14286. Springer, Cham.

Preprints

- [A3] Sacilotto, A., Bernardelli, A. M., Gualandi, S. (2026). Comparative Analysis of Linear Battery Models for Carbon Emission Optimization in Solar Energy Systems. arXiv preprint arXiv:2606.26648. Currently under revision at *Computers & Operations Research*.
- [A2] Riccardi, G., Bernardelli, A. M., Gualandi, S. (2026). Theoretical Perspectives on Jabr-Type Convex Relaxations for AC Optimal Power Flow. arXiv preprint arXiv:2604.00664. Currently under revision at *European Journal of Operational Research*.
- [A1] Bernardelli, A. M., Vercesi, E., Gualandi, S., Mastrolilli, M., Gambardella, L. M. (2024). Lower bounds for the integrality gap of the bi-directed cut formulation of the Steiner Tree Problem. arXiv preprint arXiv:2405.13773. Currently under revision at *Discrete Applied Mathematics*.

Awards

- 04/2023 Grant awarded by the TAILOR Connectivity Fund for a one-month visiting period at TU Delft, Delft, The Netherlands.

07/2022 2nd place out of 13 submissions at the 14th AIMMS-MOPTA Optimization Modeling Competition.

09/2019 Department scholarship for the Master's degree program in Mathematics.

Visiting Periods

09/2023–02/2024 I spent six months in USI (Università della Svizzera italiana), Lugano, Switzerland, as part of my Ph.D. joint program, working on integrality gap problems with Professor Luca Maria Gambardella.

04/2023 I spent one month in TU Delft, Delft, The Netherlands, as a guest of STAR Lab, working on AI and optimization with Dr. Neil Yorke-Smith.

Conference Organization

05/2025 I was one of the organizers of the COMPMAT Spring Workshop 2025, that was held at University of Pavia on May 23, 2025.

02/2025 I was part of the organizing committee of the 9th AIROYoung Workshop, that was held at University of Pavia from the 26th to the 28th of February, 2025.

2022–2024 I was a co-organizer of Caffè Beltrami, a cycle of seminars aimed at introducing different areas of research in mathematics to Bachelor's and Master's students.

Teaching Activities

Teaching

2024/2025 511180 – Algorithms for Optimization ([LM-16] Finance), University of Pavia, 22 hours. Co-teaching with Prof. Daniele Boffi, who held the other 22 hours of the course.

Supervising

2024/2025 *Riduzione delle Emissioni di Carbonio mediante Ottimizzazione Stocastica dello Stoccaggio Energetico a Batteria in Sistemi Fotovoltaici*, Anna Sacilotto, Master's Thesis in Mathematics at University of Pavia, co-supervised with Prof. Stefano Gualandi.

2024/2025 *Strengthening the SOCP formulation for the ACOPF problem*, Arthur Mazeyrat, Master's Report in Operations Research and Combinatorial Optimization at University Grenoble Alpes, co-supervised with Prof. Stefano Gualandi and Gabor Riccardi.

2023/2024 *The Cloven Travelling Salesman: a new Approach to the ATSP Integrality Gap Estimation*, Alessandro Soso, Master's Thesis in Mathematics at University of Pavia, co-supervised with Prof. Stefano Gualandi.

2022/2023 *On the exactness of Jabr-like models and distributionally robust stochastic optimal power flow*, Gabor Riccardi, Master's Thesis in Mathematics at University of Pavia, co-supervised with Prof. Stefano Gualandi.

2022/2023 *Analisi di architetture per Binarized Neural Networks*, Andrea Panno, Bachelor's Thesis in Bioengineering at University of Pavia, co-supervised with Prof. Stefano Gualandi and Simone Milanesi.

Tutoring

2021/2022 Tutoring activity for the course *Geometria e Algebra* that was held by prof. Francesco Bonsante for the Bachelor's degree program *Bioingegneria* at University of Pavia. 40 hours.

2021/2022 Tutoring activity for first-year students attending the Bachelor's degree program in Mathematics at the University of Pavia. 40 hours.

Personal Information

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